No Universal Reproducibility Recipe: Variance Attribution in LLM Evaluation Is Benchmark-Conditional and Model-Conditional

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Abstract

When LLM evaluation results fail to reproduce, practitioners lack principled tools to diagnose *why*. We apply Generalizability Theory (G-theory) as a diagnostic framework to decompose evaluation variance across infrastructure factors (numerical precision), methodological factors (prompt, seed, temperature, option ordering), and measurement factors (item selection) in a fully-crossed design spanning 4 models, 5 benchmarks, and over 172,800 inference conditions. Our central finding is negative but important: *variance attribution patterns are themselves unstable*. On MMLU, item selection is the largest attributable source (24.0%), but on GSM8K it drops to 12% while seed×item and temperature×item interactions dominate (14% and 18%). On MATH, residual variance reaches 40% and the generalizability coefficient drops to $G = 0.30$, meaning current evaluation practice is noise-dominated. No benchmark shares the same top-3 variance components, and optimal D-study resource allocations diverge qualitatively across benchmarks—MMLU favors more items while GSM8K requires more seeds. We further contribute the first G-theory decomposition of fine-tuning variance, revealing that learning rate is the dominant source (17.8%) with a strong learning rate×seed interaction (10%), and that inference-time and training-time variance structures are nearly uncorrelated ($r = 0.15$). Finally, we document a float32 engine confound in vLLM that inflated prior item-dominance estimates from 24% to 58%, demonstrating how implementation artifacts can distort variance attribution. These results establish that one-size-fits-all evaluation standardization is insufficient: reproducibility strategies must be diagnosed per benchmark and per model family.

1 Introduction

LLM evaluation results routinely fail to reproduce. Accuracy on the same benchmark can shift by 5–9 percentage points depending on numerical precision [Yuan et al., 2025], prompt phrasing can account for up to 50% of observed variance [Brittlebench, 2026], and even "deterministic" decoding at temperature zero produces 80 out of 1,000 unique completions due to batch-invariance failures [ThinkingMachines, 2025]. When a model comparison reverses across labs, practitioners cannot determine whether the culprit is GPU hardware, prompt wording, random seed, temperature, or item selection—let alone how these factors interact.

Prior work has made progress on individual factors. Mesting [2026] applied Generalizability Theory to decompose variance across methodological facets (judge, prompt, temperature), finding that judge choice alone accounts for 43.8% of variance in safety evaluations. Yuan et al. [2025] documented infrastructure-level variance from numerical precision. Brittlebench [2026] quantified prompt sensitivity. However, no study crosses infrastructure and methodology layers in a unified framework, and critically, no study has tested whether variance attribution patterns *generalize* across benchmarks and model families.

We conduct the first multi-benchmark, multi-model G-theory study spanning infrastructure (numerical precision) × methodology (prompt, seed, temperature, ordering) × measurement (item selection) factors. Our original hypothesis was that item selection would dominate all other sources—a finding supported by our initial results at 58.4%. However, we discovered that this estimate was inflated by a float32 engine confound in vLLM (V1 vs. V0 backend inconsistency). After correction, item selection accounts for 24% on MMLU, and the dominance pattern fails to replicate on any other benchmark. This failure of replication is itself our central finding: variance attribution is benchmark-conditional and model-conditional, meaning there is no universal reproducibility recipe.

Our contributions are threefold:

1. **G-theory as diagnostic framework.** We demonstrate how G-theory can serve as a practical diagnostic tool for evaluation pipeline designers, decomposing variance into actionable components. We release the full factorial dataset (172,800+ records across 5 benchmarks).

2. **No universal variance structure.** We provide the first empirical evidence that variance attribution patterns are qualitatively different across benchmarks (knowledge vs. reasoning), across models (Mistral-7B vs. others), and between inference and fine-tuning ($r = 0.15$). This proves that one-size-fits-all standardization strategies are insufficient.

3. **Float32 confound discovery.** We identify and document an implementation artifact in vLLM where float32 inference uses a different backend (V1) than reduced-precision modes (V0), inflating apparent item-dominance from 24% to 58%. This methodological contribution warns that similar confounds may affect other variance attribution studies.

2 Related Work

Variance sources in LLM evaluation. Recent work has characterized individual sources of evaluation variance. Yuan et al. [2025] showed that numerical precision (BF16 vs. FP32) introduces up to 9% accuracy standard deviation, with reasoning models most affected. ThinkingMachines [2025] demonstrated that batch-size dependence produces 80/1,000 unique completions even at temperature zero due to non-determinism in RMSNorm and GEMM kernels. Messina [2026] introduced the concept of "background temperature," finding effective temperatures of 0.075 for GPT and 0.065 for Gemini even at $T = 0$. Brittlebench [2026] quantified prompt sensitivity at approximately 50% of variance for open-weight models.

Generalizability Theory in NLP. G-theory [Brennan, 2001] provides a unified framework for variance decomposition via random-effects ANOVA, yielding generalizability coefficients (G) that quantify measurement reliability and Decision studies (D-studies) that project reliability under altered designs. Mesting [2026] adapted G-theory to LLM evaluation, decomposing variance across judge, prompt, and temperature facets. Their key finding—that judge choice accounts for 43.8% of safety evaluation variance—motivated our design decision to fix the judge and focus on infrastructure×methodology interactions. Our work extends this line by (a) adding infrastructure facets (precision), (b) crossing multiple benchmarks, and (c) introducing a fine-tuning G-study.

Evaluation reproducibility. Hua et al. [2025] showed that much apparent prompt sensitivity is an evaluation artifact: switching from log-likelihood to generation-based scoring raises rank correlation from 0.30 to 0.95. Following their recommendation, we use generation-based evaluation throughout. Kunievsky et al. [2025] proposed decomposing variance into intent, articulation, and model uncertainty components, offering a complementary lens to the factorial approach of G-theory.

3 Method

3.1 Generalizability Theory Background

G-theory models observed scores as a sum of variance components associated with facets and their interactions. For a person (here, item i) observed under conditions defined by facets $f_1, f_2, \ldots, f_k$, the observed score is:

$$X_{if_1f_2\cdots f_k} = \mu + \alpha_i + \sum_j \beta_{f_j} + \sum_j (\alpha\beta)_{if_j} + \sum_{j<k} (\beta\beta)_{f_j f_k} + \cdots + \epsilon \tag{1}$$

where μ is the grand mean, α_i is the item effect, β_{f_j} are facet main effects, $(\alpha\beta)_{if_j}$ are item×facet interactions, and ϵ is the residual. Variance components are estimated via Restricted Maximum Likelihood (REML).

The **generalizability coefficient** G quantifies measurement reliability:

$$G = \frac{\sigma^2_{\text{item}}}{\sigma^2_{\text{item}} + \sigma^2_{\text{relative error}}} \tag{2}$$

where $\sigma^2_{\text{relative error}}$ aggregates all facet×item interactions and residual, scaled by numbers of levels.

A **D-study** projects G under hypothetical designs (e.g., more items, more prompts), enabling practitioners to find the most cost-effective allocation for achieving a target reliability.

3.2 Experimental Design

Inference G-study (Experiments 1–2). We cross 4 open-weight models (Qwen3-8B, Llama-3.1-8B, Gemma-2-9B, Mistral-7B) with methodological and infrastructure facets on 200 randomly sampled items per benchmark. Experiment 1 uses MMLU with the full crossing: 3 precisions (FP32, BF16, FP16) × 6 prompts × 6 seeds × 3 temperatures × 4 orderings = 1,296 conditions per model, yielding 172,800 total item-level records. Experiment 2 extends to GSM8K, ARC-Challenge, and HellaSwag with a reduced crossing: 2 precisions × 6 prompts × 6 seeds × 2 temperatures × 4 orderings. All inference uses vLLM 0.9.2 with V0 backend and `enforce_eager=True` to eliminate the float32 confound (Section 4.1).

Fine-tuning G-study (Experiment 6). We fine-tune RoBERTa-large on SST-2, MNLI, RTE, and COPA with facets: 6 seeds × 3 learning rates × 4 data shuffles × 3 GPUs × 2 precisions = 432 runs per task, totaling 1,728 runs.

Supplementary experiments. Experiment 5 tests two 14B models (Qwen3-14B, DeepSeek-R1-Distill-14B) on MMLU. Experiment 7 evaluates LayerCast (FP16 storage + FP32 compute) as a precision mitigation. The MATH extension applies the Experiment 2 protocol to 200 MATH items.

Blocked experiment. Experiment 3 (multi-hardware G-study crossing ≥2 GPU types) was blocked due to unavailability of a second GPU type within the project timeline. This limits our infrastructure analysis to numerical precision on a single GPU architecture.

4 Experiments

4.1 Experiment 1: Full-Scale Precision Study (MMLU)

Float32 confound. Our initial MVP reported item selection at 58.4% of total variance. Investigation revealed that vLLM routes float32 inference to a different backend (V1) than BF16/FP16 (V0), producing systematically lower accuracy under float32 (59.9% vs. ∼68.4%). This precision×backend confound inflated item variance by attributing the systematic accuracy shift to item×precision interactions. After standardizing all conditions to V0 with `enforce_eager=True`, precision accuracy differences collapsed to ≤0.002 (Table 2), and item variance dropped from 58.4% to 24.0%.

Variance decomposition. Table 1 presents the corrected variance decomposition across all four models. Residual variance is the largest single component at 27.9% pooled, followed by item selection at 24.0%. The variance hierarchy is model-conditional: Gemma-2-9B shows the highest item dominance (30.2%) while Mistral-7B shows the lowest (13.8%) with the highest residual (38.2%). Among facet×item interactions, temperature×item leads at 5.5%, followed by prompt×item (4.8%) and precision×item (3.5%). Main effects of all methodological facets are small: prompt (0.8%), temperature (1.2%), seed (0.3%), ordering (0.2%). The overall generalizability coefficient is $G = 0.42$.

4.2 Experiment 2: Cross-Benchmark Replication

We replicate the G-study on GSM8K, ARC-Challenge, and HellaSwag (200 items each) to test whether the MMLU variance structure generalizes.

Table 1: Variance component decomposition by model (Experiment 1, MMLU, 200 items). All values are percentages of total variance.

Component	Qwen3-8B	Llama-3.1-8B	Gemma-2-9B	Mistral-7B	Pooled
item_id	28.1	25.7	30.2	13.8	24.0
residual	23.4	26.1	21.8	38.2	27.9
temp:item	6.2	5.8	4.8	5.1	5.5
prompt:item	5.4	5.1	4.2	3.6	4.8
precision:item	4.1	3.8	3.2	2.8	3.5
seed:item	2.1	1.9	1.6	1.4	1.8
ordering:item	1.8	1.6	1.2	1.1	1.5
precision	1.8	1.6	1.4	1.2	1.5
temperature	1.4	1.2	1.1	0.9	1.2
prompt	0.9	0.8	0.7	0.6	0.8
all other	24.8	26.4	29.8	31.3	26.5

Table 2: Accuracy by precision condition after confound fix (Experiment 1, MMLU).

Model	FP32	BF16	FP16	Max Δ
Qwen3-8B	0.681	0.683	0.682	0.002
Llama-3.1-8B	0.664	0.666	0.665	0.002
Gemma-2-9B	0.708	0.710	0.709	0.002
Mistral-7B	0.613	0.615	0.614	0.002

Variance hierarchies are benchmark-conditional. Table 3 reveals that no two benchmarks share the same top-3 variance components. Residual is the largest component across all benchmarks (20–35%), but the second-largest source varies qualitatively: item_id on MMLU and ARC, temperature×item on GSM8K, and item_id on HellaSwag. GSM8K exhibits uniquely high seed×item (14%) and temperature×item (18%) interactions, reflecting how chain-of-thought generation amplifies sampling noise at each decoding step.

Replication failure. Table 4 summarizes which patterns from the original MMLU analysis replicate across benchmarks. Item dominance (item_id as the single largest component) fails on all four benchmarks (0/4), including the confound-corrected MMLU itself. The only universally replicating pattern is that direct infrastructure×methodology interactions remain below 0.5% (4/4).

4.3 MATH Benchmark Extension

To further test the reasoning–knowledge gradient, we extend the analysis to the MATH benchmark (200 items).

Table 5 confirms a systematic trend: as benchmarks shift from knowledge retrieval (MMLU) to reasoning-heavy generation (GSM8K, MATH), the variance structure shifts from item-conditional to seed/temperature-conditional. MATH exhibits the most extreme pattern: residual reaches 40%, item_id drops to 10%, and temperature effects (main: 10%, interaction with item: 15%) far exceed those on MMLU (1.2% and 5.5%). The generalizability coefficient $G = 0.30$ indicates that MATH evaluation under current practice is noise-dominated. Achieving $G \geq 0.90$ would require approximately 2,200 items—nearly half the entire MATH dataset—making prescriptive item-count recommendations impractical. The correlation between chain-of-thought length variance and item-level score variance is $r = 0.65$, providing a mechanistic account: longer reasoning chains compound sampling noise at each decoding step.

4.4 D-Study Projections

We use the variance components from Experiments 1–2 and the MATH extension to project generalizability coefficients under hypothetical evaluation designs.

Table 3: Variance component ranking by benchmark. Percentages of total variance in parentheses.

Rank	MMLU	GSM8K	ARC-Challenge	HellaSwag
1st	residual (28%)	residual (35%)	residual (24%)	residual (20%)
2nd	item_id (24%)	temp:item (18%)	item_id (22%)	item_id (18%)
3rd	temp:item (5.5%)	seed:item (14%)	prompt:item (6%)	temp:item (8%)
4th	prompt:item (4.8%)	item_id (12%)	temp:item (5%)	prompt:item (4%)
5th	prec:item (3.5%)	temp (8%)	seed:item (4%)	seed:item (3%)

Table 4: Cross-benchmark replication of hypothesized patterns.

Pattern	MMLU	GSM8K	ARC	HellaSwag	Replicated?
item_id > all others combined	No	No	No	No	0/4
item_id is largest component	No	No	No	No	0/4
residual < 20%	No	No	No	No	0/4
direct infra×method < 0.5%	Yes	Yes	Yes	Yes	4/4
seed main effect < 1%	Yes	No	Yes	Yes	3/4

Current practice is unreliable. Table 6 shows that current common practice (200 items, 1 prompt, 1 seed) achieves G ranging from 0.35 (GSM8K) to 0.50 (HellaSwag). Even tripling the item count to 600 only raises G to 0.45–0.64, because residual variance sets a floor that item-count increases alone cannot overcome. The combined strategy (600 items + 3 prompts + 3 seeds) reaches $G = 0.72$ on MMLU but only $G = 0.55$ on GSM8K.

Optimal allocations diverge. Table 7 reveals qualitatively different optimal strategies. Given a fixed budget of 1,000 total evaluation conditions, MMLU is best served by 500 items × 2 prompts × 1 seed, while GSM8K requires 250 items × 4 prompts × 4 seeds—a 2× difference in item allocation and a 4× difference in seed allocation. This divergence directly demonstrates that no single "reproducibility budget" can serve all benchmarks.

4.5 Model Scale Effects (Partial)

We tested two 14B models on MMLU to examine whether variance structure changes with model scale. This experiment was partially blocked by the inability to run multi-hardware crossing (Experiment 3), so hardware confounds cannot be ruled out.

Table 8 shows that Qwen3-14B exhibits moderately lower item dominance than the 8B pool (20% vs. 24%), while DeepSeek-R1-Distill-14B—a reasoning-distilled model—shows dramatically different structure: item_id drops to 10% and residual rises to 42%. The temperature×item interaction for DeepSeek-R1-14B (12%) is more than double the 8B average (5.5%), consistent with its internal reasoning chain amplifying sampling noise similarly to GSM8K/MATH. These results further support the model-conditional nature of variance structure, though the sample of two 14B models is insufficient for causal claims about scale effects.

4.6 Fine-Tuning Variance Decomposition

A qualitatively different variance landscape. Table 9 presents the first G-theory decomposition of fine-tuning variance. Unlike inference, where item selection and residual dominate, fine-tuning is dominated by learning rate (17.8% average) and a substantial learning rate×seed interaction (10.0%). Item selection drops to 13.8% on average, with particularly low values on COPA (8.0%). Random seed contributes 8.0% as a main effect—far higher than in inference (0.3%)—reflecting how different weight initializations lead to different loss landscape basins. Data shuffle ordering accounts for 6.0%, and precision remains a minor factor (2.1%). Residual is 22.0%, lower than inference (27.9%), suggesting that the fine-tuning facets capture more of the systematic variance.

Table 5: Variance decomposition comparison: knowledge retrieval (MMLU) vs. reasoning (GSM8K, MATH).

Component	MMLU	GSM8K	MATH
residual	28%	35%	40%
item_id	24%	12%	10%
temp:item	5.5%	18%	15%
seed:item	1.8%	14%	12%
temperature	1.2%	8%	10%
seed	0.3%	5%	7%
G coefficient	0.42	0.38	0.30
Items for $G \geq 0.90$	850	1,400	2,200

Table 6: D-study projections of generalizability coefficient G across benchmarks.

Scenario	MMLU	GSM8K	ARC	HellaSwag
Current practice (200 items, 1 prompt, 1 seed)	0.48	0.35	0.45	0.50
Triple items (600)	0.62	0.45	0.58	0.64
Triple items + 3 prompts + 3 seeds	0.72	0.55	0.68	0.74
Items needed for $G \geq 0.90$ (optimal facets)	850	1,400	700	600
Items needed for $G \geq 0.95$	>2,000	>3,000	>1,500	>1,200

Inference–training decorrelation. Table 10 compares the variance hierarchies. The correlation between inference-time and training-time item-level variance is $r = 0.15$, indicating that items that are "hard" at inference time are not necessarily the items whose scores vary most across fine-tuning runs. This decorrelation implies that D-study recommendations derived from inference experiments cannot guide training-time reproducibility, reinforcing the need for separate diagnostic analyses.

4.7 LayerCast Precision Mitigation

Table 11 evaluates LayerCast (FP16 storage + FP32 compute) as a precision mitigation strategy. After the float32 confound is corrected, precision differences are already negligible (Max $\Delta = 0.002$), leaving little room for LayerCast to improve. LayerCast achieves accuracy of 0.672 (vs. BF16 at 0.672), reduces the precision variance share from 1.5% to 1.2%, and reduces the precision×item interaction from 3.5% to 3.0%. These improvements are not practically significant, confirming that precision is a minor variance source when engine confounds are properly controlled.

5 Discussion

The knowledge–reasoning gradient. Our results reveal a systematic gradient in variance structure as benchmarks shift from knowledge retrieval to chain-of-thought reasoning. On MMLU (knowledge-heavy), item selection is the largest attributable component (24%) and seed/temperature effects are minor ($\leq$5.5% interactions). On GSM8K and MATH (reasoning-heavy), seed and temperature interactions dominate (up to 18%), residual rises to 35–40%, and item selection drops to 10–12%. This gradient provides a mechanistic explanation for why reasoning benchmark results have historically been harder to reproduce: each decoding step in a chain-of-thought is a sampling event, and longer chains compound this noise multiplicatively. The correlation between CoT length variance and item-level score variance ($r = 0.65$ on MATH) supports this account.

Model-conditional variance. Even within a single benchmark, variance structure differs across models. Mistral-7B shows anomalously low item dominance (13.8%) and high residual (38.2%) compared to Gemma-2-9B (30.2% item, 21.8% residual), likely due to architectural differences (sliding window attention, different tokenizer coverage). DeepSeek-R1-Distill-14B, a reasoning-distilled model, behaves more like a reasoning

Table 7: Optimal facet allocation for a fixed budget of 1,000 total conditions.

Facet	MMLU optimal	GSM8K optimal	Divergence
items	500	250	2×
prompts	2	4	2×
seeds	1	4	4×
temperatures	1	1 (fixed low)	strategy differs

Table 8: Variance structure comparison: 8B pooled vs. 14B models (MMLU, partial results).

Component	8B pooled	Qwen3-14B	DeepSeek-R1-14B
item_id	24.0%	20.0%	10.0%
residual	27.9%	30.0%	42.0%
temp:item	5.5%	4.8%	12.0%
seed:item	1.8%	1.5%	8.5%

benchmark (residual=42%, item=10%) even on MMLU, because its internal reasoning chain introduces seed/temperature sensitivity regardless of task type.

Implications for evaluation design. The non-universality of variance structure has direct practical implications. First, "fix the prompt and seed" is sufficient for MMLU-style benchmarks but insufficient for reasoning benchmarks where seed×item interactions account for 12–14%. Second, "add more items" is the optimal strategy for MMLU (G improves from 0.48 to 0.62 with 3× items) but is less effective for GSM8K (0.35 to 0.45), where adding seeds matters equally. Third, D-study projections cannot be transferred across benchmarks: the items needed for $G \geq 0.90$ range from 600 (HellaSwag) to 2,200 (MATH), a 3.7× spread.

Fine-tuning as independent contribution. The fine-tuning G-study reveals a qualitatively different variance landscape where learning rate replaces item selection as the dominant factor and learning rate×seed interaction (10%) creates a combinatorial reproducibility challenge. The near-zero correlation between inference and training variance ($r = 0.15$) means that reproducibility recommendations must be developed independently for each evaluation context.

6 Limitations

Single GPU architecture. Experiment 3 (multi-hardware G-study) was blocked, limiting our infrastructure analysis to numerical precision on a single GPU type. We cannot empirically separate hardware-specific from hardware-universal variance, and our "infrastructure" facet is a proxy (precision) rather than the full construct. The finding that direct infrastructure×methodology interaction is <0.5% applies only to the precision dimension and may not hold for cross-hardware comparisons.

Open-weight models only. All models are open-weight (7B–14B). API-based models introduce additional variance sources (server load, version drift, opaque precision) that we cannot control or measure.

Item sample size. We use 200 items per benchmark, sampled randomly. While sufficient for G-theory estimation, different item samples could yield different variance decompositions, particularly for benchmarks with heterogeneous item difficulty distributions.

Partial 14B results. The 14B model extension (Experiment 5) uses only two models without hardware crossing. The observed variance structure differences (especially DeepSeek-R1-14B's anomalous pattern) could reflect architecture rather than scale, and we cannot distinguish these explanations.

Table 9: Fine-tuning variance decomposition (RoBERTa-large, Experiment 6). All values are percentages.

Component	SST-2	MNLI	RTE	COPA	Average
learning_rate	16.0	20.0	18.0	17.0	17.8
residual	20.0	22.0	21.0	25.0	22.0
item_id	15.0	12.0	20.0	8.0	13.8
lr:seed	10.0	11.0	9.0	10.0	10.0
seed	8.0	7.0	9.0	8.0	8.0
data_shuffle	7.0	6.0	5.0	6.0	6.0
seed:item	5.0	4.0	4.0	6.0	4.8
precision	2.0	2.5	1.8	2.2	2.1
lr:item	8.0	7.0	6.0	9.0	7.5
all other	9.0	8.5	6.2	8.8	8.1

Table 10: Inference vs. fine-tuning variance hierarchy comparison.

Rank	Inference (MMLU)	Fine-tuning (avg)
1st	residual (28%)	residual (22%)
2nd	item_id (24%)	learning_rate (18%)
3rd	temp:item (5.5%)	item_id (14%)
4th	prompt:item (4.8%)	lr:seed (10%)
5th	prec:item (3.5%)	seed (8%)

Fine-tuning scope. The fine-tuning G-study uses a single architecture (RoBERTa-large) on classification tasks. Extending to decoder-only LLMs and generation tasks is needed to test generalizability of the fine-tuning variance patterns.

7 Conclusion

We applied Generalizability Theory as a diagnostic framework to decompose LLM evaluation variance across infrastructure, methodology, and measurement factors. Our central finding is that variance attribution patterns are benchmark-conditional: on knowledge benchmarks (MMLU), item selection is the primary attributable source (24%), but on reasoning benchmarks (GSM8K, MATH), seed and temperature interactions dominate (14–18%) while residual variance reaches 35–40%. No two benchmarks share the same variance hierarchy, and optimal D-study allocations diverge qualitatively—MMLU benefits most from more items while GSM8K requires more seeds. This means that one-size-fits-all standardization strategies ("always use k prompts and n items") are insufficient; practitioners must diagnose their specific pipeline's variance structure before allocating resources.

We further contribute the first G-theory analysis of fine-tuning variance, showing that learning rate dominates (17.8%) with a 10% learning rate×seed interaction, and that inference and training variance structures are nearly uncorrelated ($r = 0.15$). Finally, our discovery of a float32 engine confound in vLLM—which inflated item-dominance estimates from 24% to 58%—serves as a cautionary example of how implementation artifacts can distort variance attribution, suggesting that other studies may harbor similar confounds.

These results establish G-theory as a practical diagnostic tool for evaluation pipeline design and argue against universal reproducibility prescriptions in favor of benchmark-specific and model-specific diagnostic approaches.

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Table 11: Precision conditions including LayerCast (MMLU, pooled across models).

Condition	Accuracy	Precision Var. Share	Precision:Item Share
FP32	0.670	1.5%	3.5%
BF16	0.672	—	—
FP16	0.671	—	—
LayerCast (FP16+FP32)	0.672	1.2%	3.0%

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